

Chapter 16

Outlook: Sparsity of the Ternary Collision Tensor

In Section 15 we noted that applying the irreducible Burnett framework developed in this thesis to collision operators of a different kind – including higher-order interactions – looks like a promising next step. In this chapter we take a first step in that direction, by showing that the sparsity mechanism underlying the entire approach of this thesis is not special to binary collisions: it extends, with no new theoretical machinery, to multi-particle (in particular ternary) collision operators satisfying the same rotational equivariance. We state and prove the relevant selection rule and quantify the resulting sparsity at the level of degrees of freedom (Section 16.5), and we close with a concrete account of what remains to make this computational (Section 16.6), which we leave to future work.

16.1 Motivation: a ternary extension of the Boltzmann equation

A natural ternary extension of the Boltzmann equation, modeling three-particle interactions relevant for denser gases, has recently been studied rigorously by Ampatzoglou, Gamba, Pavlović and Tasković [1], building on the rigorous derivation of the ternary collision operator by Ampatzoglou and Pavlović [2]. There, the binary-ternary Boltzmann equation is written

$$\partial_t f = Q_2(f, f) + Q_3(f, f, f),$$

with Q_3 a trilinear collision operator built from an explicit, rotationally invariant collisional law analogous to (3.1)–(3.3) of this thesis, but coupling three particles at once. We ask the question motivating this chapter: if such a Q_3 is expressed in the same basis used throughout this thesis – adapted to the O_3 -irreducible decomposition of the moment space – does its collision tensor exhibit the same kind of sparsity as the binary collision tensor of Chapter 10?

We answer this in the affirmative.

16.2 The trilinear natural map and its selection rule

Consider a trilinear map $Q : \mathcal{P}^M \times \mathcal{P}^M \times \mathcal{P}^M \rightarrow \mathcal{P}^M$ satisfying $Q(RF, RG, RH) = RQ(F, G, H)$ for all $R \in O_3$, the direct trilinear analogue of the bilinear naturality condition used throughout Chapters 7–8. As in Section 8.4, restricting to a quadruple of irreducible blocks at a time, the relevant object is

$$\mu(n_0, n_1, n_2, n_3) := \dim \operatorname{Hom}_{O_3}(H^{n_1} \otimes H^{n_2} \otimes H^{n_3}, H^{n_0}).$$

Theorem 7.2.13 (Brauer’s classification) applies unchanged here, since it classifies O_3 -natural maps $V^{\otimes n} \rightarrow V^{\otimes m}$ for arbitrary n, m , not only the bilinear case $n = 2, m = 1$ used in Chapter 8. What changes relative to Lemma 8.4.1 is the multiplicity: unlike the bilinear case, μ need not be 0 or 1.

Theorem 16.2.1 (Trilinear selection rule and multiplicity). $\mu(n_0, n_1, n_2, n_3) = 0$ unless $n_0 + n_1 + n_2 + n_3$ is even, in which case

$$\mu(n_0, n_1, n_2, n_3) = \max\left(0, \min(n_1 + n_2, n_3 + n_0) - \max(|n_1 - n_2|, |n_3 - n_0|) + 1\right). \quad (16.1)$$

Moreover, μ is invariant under all permutations of (n_1, n_2, n_3) , and, using self-duality of H^{n_0} , under all permutations of (n_0, n_1, n_2, n_3) .

Proof. Parity. $O_3 \cong SO_3 \times \{I, -I\}$, with $-I$ central of order 2. By the argument of Lemma 7.3.3 (the polynomial harmonic space H^n satisfies $(-I) \cdot p = (-1)^n p$, since each of the three variables changes sign under $x \mapsto -x$), $-I$ acts on $H^{n_1} \otimes H^{n_2} \otimes H^{n_3}$ as the scalar $(-1)^{n_1+n_2+n_3}$. A nonzero O_3 -equivariant map to H^{n_0} must intertwine this action, forcing $(-1)^{n_1+n_2+n_3} = (-1)^{n_0}$, i.e. $n_0 + n_1 + n_2 + n_3$ even – the direct generalization of the parity condition in (8.8).

Multiplicity. As SO_3 -representations, the Clebsch–Gordan series gives the multiplicity-free decomposition $H^{n_1} \otimes H^{n_2} \cong \bigoplus_{J=|n_1-n_2|}^{n_1+n_2} V_J$. For each such J , the multiplicity of V_{n_0} in $V_J \otimes H^{n_3}$ is 1 if $|J - n_3| \leq n_0 \leq J + n_3$ and 0 otherwise, again by Clebsch–Gordan. The total SO_3 -multiplicity of V_{n_0} in the iterated product is therefore the number of integers J lying in both $[|n_1 - n_2|, n_1 + n_2]$ and $[|n_0 - n_3|, n_0 + n_3]$, giving (16.1). Since $-I$ acts as the single scalar $(-1)^{n_1+n_2+n_3}$ on the entire ambient space, it acts identically on every SO_3 -summand; given the parity condition holds, this eigenvalue matches that of the genuine H^{n_0} (rather than its parity-twisted partner), so the SO_3 -multiplicity computed above equals the O_3 -multiplicity μ .

Permutation invariance follows since μ is, by definition, the dimension of the space of O_3 -invariant vectors in $H^{n_0} \otimes H^{n_1} \otimes H^{n_2} \otimes H^{n_3}$ (using that every H^n is self-dual), manifestly independent of the order of the four factors. $\square$

Remark 16.2.2. Setting $n_3 = 0$ recovers Lemma 8.4.1 exactly: with $n_3 = 0$, (16.1) reduces to $\mu = 1$ on the locus $|n_1 - n_2| \leq n_0 \leq n_1 + n_2$ and 0 off it, recovering the binary selection rule (8.8) as the two-input special case of this three-input statement.

Remark 16.2.3 (General k -ary interactions). Nothing in the proof of Theorem 16.2.1 used that there are exactly three input factors. The same two-step argument – a single global parity obstruction from the action of $-I$, together with an iterated Clebsch–Gordan count of surviving intermediate channels – applies verbatim to a k -linear natural map $\mathcal{P}^{M \times k} \rightarrow \mathcal{P}^M$ for any $k \geq 2$, with the multiplicity $\mu(n_0, n_1, \dots, n_k)$ obtained by $(k - 1)$ -fold iteration of the binary Clebsch–Gordan rule. We do not pursue the general statement here, but record this as the natural further generalization of the present chapter.

We independently verified Theorem 16.2.1 by an unrelated method: numerically evaluating the SO_3 character integral

$$\int_0^\pi \chi_{n_0}(\theta) \chi_{n_1}(\theta) \chi_{n_2}(\theta) \chi_{n_3}(\theta) \frac{2}{\pi} \sin^2(\theta/2) d\theta, \quad \chi_l(\theta) = \frac{\sin((l + \frac{1}{2})\theta)}{\sin(\theta/2)},$$

against formula (16.1) for a battery of test quadruples, including high-multiplicity cases (e.g. $(5, 5, 5, 5) \mapsto \mu = 11$). Agreement was exact in every even-parity case tested, and odd-parity quadruples confirmed that the parity condition discards genuine nonzero SO_3 -level couplings, as the proof predicts.

16.3 Reduction to a totally symmetric map

Exactly as in the binary case (cf. Table 10.3.1, Case 6), since the physical equation only ever evaluates Q on the diagonal $Q(F, F, F)$, we may assume Q totally symmetric without loss of generality.

Lemma 16.3.1 (Polarization). *For any trilinear map $Q : \mathcal{P}^{M \times 3} \rightarrow \mathcal{P}^M$, the diagonal value $Q(F, F, F)$ depends only on the totally symmetric part $Q_{\text{sym}}(F, G, H) := \frac{1}{3!} \sum_{\sigma \in S_3} Q(F_{\sigma(1)}, F_{\sigma(2)}, F_{\sigma(3)})$.*

Proof. Write $Q = Q_{\text{sym}} + Q_{\text{rem}}$, where $\sum_{\sigma \in S_3} Q_{\text{rem}}(F_{\sigma(1)}, F_{\sigma(2)}, F_{\sigma(3)}) \equiv 0$. Restricting to $F_1 = F_2 = F_3 = F$ collapses this sum to $3! \cdot Q_{\text{rem}}(F, F, F) = 0$, so $Q(F, F, F) = Q_{\text{sym}}(F, F, F)$. $\square$

16.4 Conservation constraints

By the argument of Section 12.8, conservation of mass, momentum and energy forces every surviving channel of the ternary impact tensor contributing to output block $(n_0, a_0) \in \{(0, 0), (1, 0), (0, 1)\}$ to vanish – now for *every* multiplicity index in $\{1, \dots, \mu(n_0, n_1, n_2, n_3)\}$ in that channel, rather than the single coefficient of the binary case. This is, as in Chapter 12, a physical constraint imposed in addition to Theorem 16.2.1, not a consequence of it.

16.5 Sparsity of the ternary collision tensor

Generalizing the Main Theorem 8.4.2, any O_3 -natural trilinear map decomposes as

$$Q(F_1, F_2, F_3) = \sum_{\substack{n_0, n_1, n_2, n_3 \\ a_0, a_1, a_2, a_3}}^{\mu(n_0, n_1, n_2, n_3)} \sum_{\mu=1} S_{a_0 a_1 a_2 a_3, \mu}^{n_0 n_1 n_2 n_3} \cdot \Phi_{a_0 a_1 a_2 a_3, \mu}^{n_0 n_1 n_2 n_3}(F_1, F_2, F_3),$$

with Φ a fixed, universal basis of natural maps for the surviving channel and S the physics-dependent impact tensor. The non-symmetrized true degree-of-freedom count is

$$\text{DOF}_3(M) = \sum_{n_0, n_1, n_2, n_3=0}^M \text{mult}(n_0, M) \text{mult}(n_1, M) \text{mult}(n_2, M) \text{mult}(n_3, M) \mu(n_0, n_1, n_2, n_3),$$

with mult the isotropic multiplicity of (9.1)/(16.2). Table 16.1 reports this against the naive count $L(M)^4$, alongside the analogous binary quantities of Table 10.3.1 for comparison; all binary-case columns reproduce Table 10.3.1 exactly.

$$\text{mult}(n, M) = \left\lfloor \frac{M-n}{2} \right\rfloor + 1, \quad L(M) = \sum_{n=0}^M \text{mult}(n, M)(2n+1). \quad (16.2)$$

Remark 16.5.1. At $M = 5$, the ternary tensor's true DOF is 0.14% of its naive size L^4 , versus 0.28% for the binary tensor relative to L^3 – the ternary tensor is, proportionally, sparser still. This is consistent with the mechanism of Theorem 16.2.1: each additional input multiplies the domain's combinatorial label space by a further factor of $O(M)$, while the codomain $\mathcal{P}^M$ offers only $O(M)$ labels regardless of how many inputs feed into it; the multiplicity μ that compensates for the loss of Schur uniqueness grows only linearly in the indices, not enough to offset the additional triangle inequality contributed by each extra leg.

Table 16.1: Sparsity of the binary vs. ternary collision tensor (non-symmetrized impact tensor).

M	$L(M)$	L^3	binary true DOF	L^4	ternary true DOF	ternary % of L^4
1	4	64	4	256	10	3.91%
2	10	1,000	24	10,000	122	1.22%
3	20	8,000	72	160,000	746	0.47%
4	35	42,875	213	1,500,625	3,687	0.25%
5	56	175,616	486	9,834,496	13,884	0.14%

16.6 Remarks on a computational path forward

We close by recording what we view as the natural next step, which we leave open here. Producing an explicit numerical algorithm in the spirit of Chapter 12 requires computing the ternary analogue of the scattering vector integral, i.e. explicitly evaluating Φ against an actual collisional law such as that of [1]. We record one structural observation relevant to that undertaking.

The exact computability of the binary scattering integral in Chapter 12 rests on Lemma 12.1.1: the stabilizer in O_3 of the (single) relative-velocity direction $\hat{u}$ is the continuous group O_2 , which collapses the scattering integral to a function of one scattering angle. The ternary collisional law of [1] is built from *two* relative-velocity directions, $\hat{u}^{(1)} = (v_1 - v)/|\tilde{u}|$ and $\hat{u}^{(2)} = (v_2 - v)/|\tilde{u}|$.

Proposition 16.6.1. *For generic (non-collinear) v, v_1, v_2 , the stabilizer $\{R \in O_3 : R\hat{u}^{(1)} = \hat{u}^{(1)}, R\hat{u}^{(2)} = \hat{u}^{(2)}\}$ is the two-element group generated by the reflection through $\text{span}(\hat{u}^{(1)}, \hat{u}^{(2)})$, rather than a continuous group.*

Proof. If $\hat{u}^{(1)}, \hat{u}^{(2)}$ are linearly independent, spanning a plane $P \subset \mathbb{R}^3$, any R fixing both vectors fixes P pointwise and acts on the orthogonal complement $P^\perp$ (one-dimensional) as an element of $O(1) = \{\pm 1\}$, giving exactly the identity and the reflection across P . $\square$

Consequently the ternary scattering integral does not collapse to a one-variable function the way the binary one does – a finite stabilizer constrains only an even/odd parity, not the full dimensional reduction a continuous stabilizer provides. We expect the practical route to an explicit ternary impact tensor to be numerical (quadrature or Monte Carlo evaluation over the residual two-parameter shape moduli of the pre-collision configuration, at the low truncation orders where Theorem 16.2.1 guarantees only a small number of channels survive) rather than the closed-form recursion available in the binary case. We leave this computation, and the resulting numerical implementation, to future work.

Bibliography

- [1] I. Ampatzoglou, I. M. Gamba, N. Pavlović, M. Tasković, *Moment estimates and well-posedness of the binary-ternary Boltzmann equation*, preprint, 2026.
- [2] I. Ampatzoglou, N. Pavlović, *Rigorous derivation of a ternary Boltzmann equation for a classical system of particles*, Comm. Math. Phys. 387 (2021), no. 2, 793–863.